# Lecture 11 recap

In Lecture 11:

- Learnt the existence of the inverse Laplace transform and its linear properties
- Learnt to calculate LTs and ILTs using the formulae "cheat" sheet + linearity
- Learnt to use partial fractions to calculate ILTs of suitable functions

# Lecture 12

In Lecture 12, we will:

1. Learn how to express the LT of derivatives in a convenient form
2. Learn how to use the LT to solve IVPs
3. Learn the $s$-shifting and $t$-shifting theorems to calculate LTs and ILTs

## Theorem (Laplace transform of the derivative)

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$$= \lim_{b\to\infty} \left\{ \left[\mathrm{e}^{-st} f(t)\right]_{\cancel{0}\ t=0}^{b \to 0} - \int_0^b -s\,\mathrm{e}^{-st}\, f(t)\ \mathrm{d}t \right\}$$

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&= \lim_{b\to\infty} \left\{ \Big[\mathrm{e}^{-st} f(t)\Big]_0^b - \int_0^b -s\,\mathrm{e}^{-st}\, f(t)\ \mathrm{d}t \right\} \\
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as required

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$$\mathscr{L}^{-1}\{Y\} = \mathscr{L}^{-1}\left(\frac{1}{s-1}\right) = \mathrm{e}^{t} \quad \text{by LT table}$$

## Theorem (Higher order derivatives)

Let $f, f', \ldots, f^{(n-1)}$ be continuous on $[0, \infty)$ and have Laplace transforms. Then

$$\mathscr{L}\left\{f^{(n)}\right\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{(n-1)}(0)$$

Can be proved using integration by parts and induction.

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Can be proved using integration by parts and induction.

We will use the case of a second derivative extensively:

$$\mathscr{L}\{f''\} = s^2 F(s) - s f(0) - f'(0)$$

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Therefore

$$Y = \frac{2}{s^2 - 3s + 2}$$

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Therefore

$$Y = \frac{2}{s^2 - 3s + 2} = \frac{2}{(s-1)(s-2)}$$

$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = 2(\mathrm{e}^{2t} - \mathrm{e}^{t}) \qquad \text{(Ex 3.10)}$$

**Exercise 3.13**: Find the Laplace transform of $\mathrm{e}^{3\,t}\,t^2$

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**Theorem ($s$-shift)**

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega\,t} f(t)\} = F(s-\omega), \qquad \mathrm{Re}(s) > \alpha + \omega$$

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$$\mathscr{L}^{-1}\left(\frac{3}{s^2+9}\right) = \sin 3t \qquad \text{from LT table}$$

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**Exercise 3.14**: Find the inverse Laplace transform of $\dfrac{3}{(s+2)^2+9}$

$\mathcal{L}(\sin 3t) = 3/(s^2+9)$ $\quad \mathrm{Re}(s) > 0$

$$\mathscr{L}^{-1}\left(\frac{3}{s^2+9}\right) = \sin 3t \qquad \text{from LT table}$$

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Where is the shifted transform valid? $\mathrm{Re}(s) > -2$

# Discontinuous functions

**Definition (Heaviside step function)**

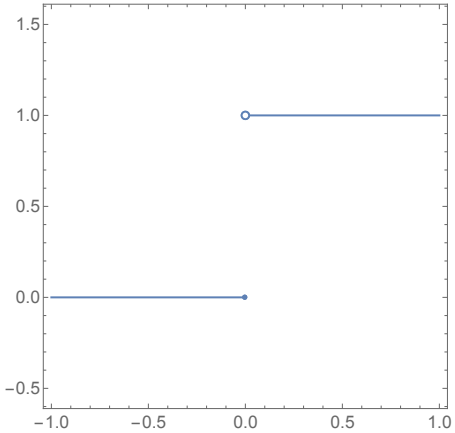
$$H(t) = \begin{cases} 0 & t \leqslant 0 \\ 1 & t > 0 \end{cases}$$

The value of $H(0)$ is usually irrelevant.

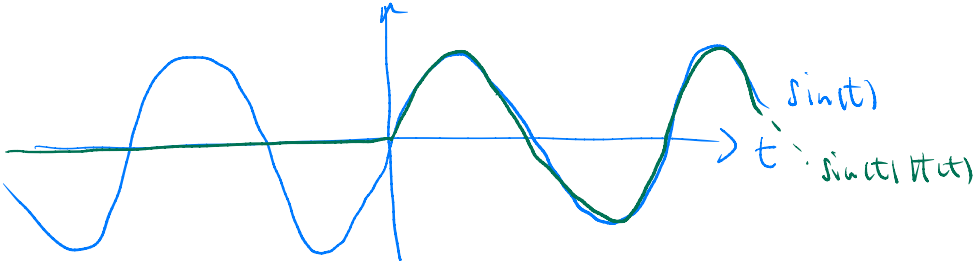
It can be translated so the discontinuity occurs at $t = c$, i.e.,

$$H(t - c)$$

It can be used to model turning processes on or off.



## **Exercise 3.15**: Find the Laplace transform of $f = \sin(t-3)H(t-3)$



## **Exercise 3.15**: Find the Laplace transform of $f = \sin(t-3)H(t-3)$

**Theorem ($t$-shift)**

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{f(t-c)H(t-c)\Big\} = \mathrm{e}^{-cs}F(s), \quad \mathrm{Re}(s) > \alpha$$

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$$F(s) = \mathrm{e}^{-3\overset{s}{\cancel{t}}}\mathscr{L}(\sin t)$$

$$\Rightarrow \quad F(s) = \frac{\mathrm{e}^{-3s}}{s^2+1} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

**Exercise 3.16**: Find the ILT of $\dfrac{\mathrm{e}^{-2s}}{s-4}$ where $\mathrm{Re}(s) > 4$

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$$\mathscr{L}^{-1}\left\{\frac{1}{s-4}\right\} = \mathrm{e}^{4t} \quad \text{for} \quad \mathrm{Re}(s) > 4$$

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$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s-4}\right\} = \mathrm{e}^{4(t-2)}H(t-2) \quad \text{for} \quad \mathrm{Re}(s) > 4$$